Duong Ta

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Last updated: October 27, 2026

Fields & Research Focus

Primary: Applied Econometrics; Agricultural Economics **Secondary:** Environmental and Resource Economics

I study how climate shocks, water scarcity, infrastructure, and policy incentives affect land-use decisions, agricultural outcomes, and environmental adaptation. My work applies causal-inference methods, including difference-in-differences, instrumental variables, and double machine learning, to household survey, panel, and spatial data.

Education

- **Virginia Tech** *2023 – Present*
  Ph.D. in Agricultural and Applied Economics USA
- **University of Greenwich** *2014 – 2015*
  MSc in Business and Financial Economics United Kingdom
- **National Economics University** *2004 – 2008*
  Bachelor in Economics Vietnam

References

Bradford Mills
Professor
Agricultural and Applied Economics
Virginia Tech
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Mario Ortez
Assistant Professor of Agribusiness
Agricultural and Applied Economics
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Kurt Stephenson
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Job Market Paper

"Estimating the Economic Impacts of Climate Stress on U.S. Dairy Production: Evidence from Debiased Machine Learning and Simulation-Based Estimator Choice"

Duong Ta, Mario Ortez, Bradford Mills, and Michael Cary. *Job market paper. Working paper.*
Links monthly milk yield and fat content in 21 major U.S. dairy states (2003–2025) to hourly ERA5-Land weather. Uses synthetic-data simulations to select a cubic regression for heat and a debiased neural network for cold. Estimates a 0.29% cumulative yield reduction per additional 24-hour-equivalent day at a temperature-humidity index of at least 69, over the exposure month and three lags. The cold estimate's 95% interval includes zero, and magnitudes depend on the assumed response function.

Publications

Journal Articles in English:

[1] Bradford Mills, Chi Ta, **Duong Phuc Ta**, Huong Nguyen (2025). "Salinity Inundation, Profitability, and Rice Farming Exits in Mekong Delta of Vietnam". *Journal of the Agricultural and Applied Economics Association*, 4(2), pp. 170–186. DOI: 10.1002/jaa2.70004

[2] Bradford Mills, Dung Phuong Le, **Duong Phuc Ta**, Lien Nhu, Danh Thanh Vo and Ricardo Labarta (2022). "Intensive and extensive rice farm adaptations in salinity-prone areas of the Mekong Delta". *Climate and Development*, 15(2), pp. 162–176. DOI: 10.1080/17565529.2022.2072800

[3] Nguyen Dinh Chuc and **Ta Phuc Duong** (2019). "Trade, Structural Adjustments and Productivity Growth in Vietnam: The Shift to Services". *Journal of Southeast Asian Economies*, 36(2), pp. 256–273.

[4] **Ta Phuc Duong** (2015). "Effects of Capital Flows on Housing Prices in Emerging Economies: A Panel Approach". *Vietnam's Socio-Economic Development*, No. 81 (April 2015).

Journal Articles (in Vietnamese) *(published under the name Ta Phuc Duong)*:

[1] **Ta Phuc Duong** (2023). "Welfare impacts of non-farm livelihood participation of households in Vietnamese rural area". *Journal of Economic Studies*, 12(523).

[2] **Ta Phuc Duong** (2021). "Impacts of agricultural portfolio changes on farm household income under the context of climate changes in Vietnam Mekong Delta". *Journal of Economic Studies*, 12(523).

[3] **Ta Phuc Duong** and Nguyen Thi Anh Tuyet (2018). "Trends in economic functions of Vietnamese households". *Journal of Economic Studies*, 7(482).

[4] **Ta Phuc Duong** (2017). "Regional and industrial specialization of labour in Vietnam". *Journal of Regional Sustainable Development*, 12.

Working Papers

[1] **Duong Ta**, Bradford Mills, Kurt Stephenson, John Fike. "Economic Feasibility and Carbon Sequestration of Silvopasture in the Southeastern U.S." *Submitted to a journal and under review.*

Research in Progress

[1] **Duong Ta** (first author). "Server Farms and Agricultural Land: Data-Center Development, Farmland Prices, and Land-Market Activity in Virginia." *Work in progress.*

Conference Presentations

[1] **Duong Phuc Ta**, Bradford Mills, Kurt Stephenson, John Fike, Tom Stanley (2025). "Economic Feasibility and Carbon Sequestration of Silvopasture in the Southeastern U.S." *Presented at* the AAEA-WAEA Annual Meeting, July 27–29, 2025, Denver, CO.

[2] Nguyen Chien Thang and **Ta Phuc Duong** (2019). "Changing trend of production function of Vietnamese households". RC06-VSA International Conference: The Family in Modern and Global Societies: Persistence and Change. October 17–19, 2019, Hanoi, Vietnam.

Work Experience

- **Virginia Tech [🌐]** *2023 – present*
  Graduate Research Assistant USA
  - Research for The Dairy Alliance-funded project "Enhancing dairy sector resilience through data-driven infrastructure planning and forecast modeling," including the data construction and estimation in the job market paper.
- **The International Center for Tropical Agriculture (CIAT), Asia Hub [🌐]** *2018 – 2022*
  Research Assistant / Data Analyst (held concurrently with the VIE research post) Vietnam
  - Contributed to salinity-intrusion impact assessment in the Mekong Delta (2019–2022) and cassava production and value-chain research in Vietnam (2019–2021).
  - Developed questionnaires and programmed CSPro data-entry applications for two household surveys (rice farming under salinity; cassava production), each covering about 1,000 farm households.
  - As field supervisor, led roughly three-day trainings for teams of about 12 enumerators, supervised data collection, cleaned and analyzed the data, and contributed to co-authored research.
- **Vietnam Institute of Economics (VIE) [🌐]** *2010 – 2023*
  Researcher Vietnam
  - Led projects on the dynamics of farm and non-farm employment in Vietnam's rural economy (2022) and agricultural structure under climate change in the Mekong Delta (2021), from proposal development through analysis and final reports.
  - Contributed to research on household economic functions, labour productivity, structural transformation, agricultural value chains, and farm-firm linkages.

TEACHING, TRAINING & MENTORING EXPERIENCE

- **Virginia Tech [🌐]** *2023 – present*
  Graduate Teaching Assistant USA
  - Master's Microeconomics: responsibilities primarily involved grading.
- **Vietnam Institute of Economics**
  Workshop Instructor, Introduction to Stata Vietnam
  - Designed and taught a three-session workshop for researchers on data preparation, analysis, and reproducible workflows.
  - Covered installation, cleaning, reshaping, summaries, and visualization. The final mini-project included a do-file, table, figure, and short memo.
- **International Center for Tropical Agriculture (CIAT)** *2019 – 2022*
  Field Supervisor and Enumerator Trainer Vietnam
  - Led roughly three-day trainings for teams of about 12 enumerators on the questionnaire and CSPro data-entry application before each of two household surveys, then supervised fieldwork.

PROFESSIONAL SERVICE & AFFILIATIONS

Affiliations: Agricultural and Applied Economics Association (AAEA).

SKILLS

- **Statistical & Econometric Software:** Stata, R, Python, Matlab, SQL, LaTeX
- **Geospatial & Remote Sensing:** Google Earth Engine
- **Machine Learning & Data Science:** Scikit-learn, TensorFlow, Keras
- **Reproducibility & Collaboration:** Git, GitHub, Quarto, VSCode
- **Research Methods:** Econometrics, Causal Inference, Microeconomic Methods, Cost-Benefit Analysis, Machine Learning, Survey Design, Experimental Design, Data Management, Data Visualization
- **Additional Training:** Machine Learning Specialization, Stanford University via Coursera (3 courses), 2025

LANGUAGES

Vietnamese (native), English (fluent).